MILITARY SPECIFICATION

TRANSMITTING SET, TELEMETRIC DATA

AN/DKT-59

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, development, and test requirements for the AN/DKT-59, Transmitting Set, Telemetric Data, critical item, referred to herein as the transmitting set.

1.2 Classification. The transmitter sets covered by this specification are classified as specified (see 6.2.1):

Type I - 225 Megahertz (MHz) to 260 MHz

Type II - 1435 MHz to 1540 MHz

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standarization Department (SESD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5815

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-C-85651	Converter, Voltage to Frequency CV-3746/DKT-59
MIL-A-85652	Amplifier, Mixer AM-7157/DKT-59
MIL-T-85653	Transmitter, Radio T-1458/DKT-59
MIL-P-85654	Power Supply PP-7816/DKT-59

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

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MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-2078	Requirements for Preparation of Support Equipment Depot Level Rework Standards
MIL-STD-45662	Calibration System Requirements

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Indent 30003)

1356AS309	Transmitter Set, Telemetric Data

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INTER RANGE INSTRUMENTATION GROUP (IRIG) STANDARD 106-80 — Telemetry Standards

(Application for copies should be addressed to the Department of Defense Technical Information Center, Cameron Station, Alexandria, VA 22314.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Transmitting set description. The transmitting set is an assembly of major components (see 3.1.1) integrated to form a telemetering system designed to monitor and process flight performance of the BQM-74C, BQM-34A, -34E, and other aircraft targets referred to herein as the vehicle. The transmitting set measures quantities and resultant functions of the vehicle and transmits that data in accordance with this specification. The transmitting set will be used during pre-flight and mission flight to provide data for technical and test evaluation. Operation of the transmitting set shall not affect other vehicle subsystem or system operation.

3.1.1 Transmitting set configuration. Figure 1 shows the baseline configuration of the transmitting set. Major components of the transmitting set shall be: 10 each voltage-to-frequency converters, a mixer amplifier module, a transducer power supply module, a mounting plate with attached module assembly unit (mount), a radio transmitter module, and interface cabling. Detailed definition for each module is provided in the applicable component specification listed below:

a.	MIL-C-85651	Converter, Voltage-to-Frequency, CV-3746/DKT-59
b.	MIL-A-85652	Amplifier, Mixer, AM-7157/DKT-59

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c. MIL-T-85653 Transmitter, Radio T-1458/DKT-59

d. MIL-P-85654 Transducer Power Supply, PP-7816/DKT-59

3.1.2 Physical characteristics. The transmitting set physical characteristics shall be in accordance with Drawing 1356AS309.

3.1.3 Interface definition. The inter-system interfaces (transmitting set and the vehicle) and the intra-system interfaces (between major components within the transmitting set) shall be as shown in Drawing 1356AS309. The electrical interface for the transmitting set and the vehicle is provided through connectors J101 and J102 as shown in figure 1. The cabling, connector P101 and P102, shall be installed at the Government using agency. The transmitting set must operate in a stable non-oscillitory mode when subjected to possible capacitive, inductive and resistive source signal components. Data functions originating in the vehicle shall be buffered from source impedances to prevent damaging operation to the vehicle. The intra-system interfaces between major components of the transmitting set shall be compatible with the detailed descriptions provided in the applicable component specifications (see 3.1.1).

3.1.4 Components. The transmitting set shall consist of the following major components:

a. Voltage-to-frequency converter

b. Mixer amplifier module

c. Radio transmitter module

d. Transducer power supply module

3.1.5 Government-furnished property. Unless otherwise specified in the contract or purchase order (see 6.2.1), Government-furnished property is not applicable to this specification.

3.2 Characteristics.

3.2.1 Performance. Unless otherwise specified in the contract or purchase order (see 6.2.1), the transmitting set shall meet the performance requirements specified herein under the conditions of 3.2.6.

3.2.1.1 External Power. The transmitting set shall operate using external power from the vehicle with characteristics as specified in MIL-STD-704, and modified herein as follows (see 4.6.2.2):

a. 24.0 to 32.0 volts direct current (Vdc)

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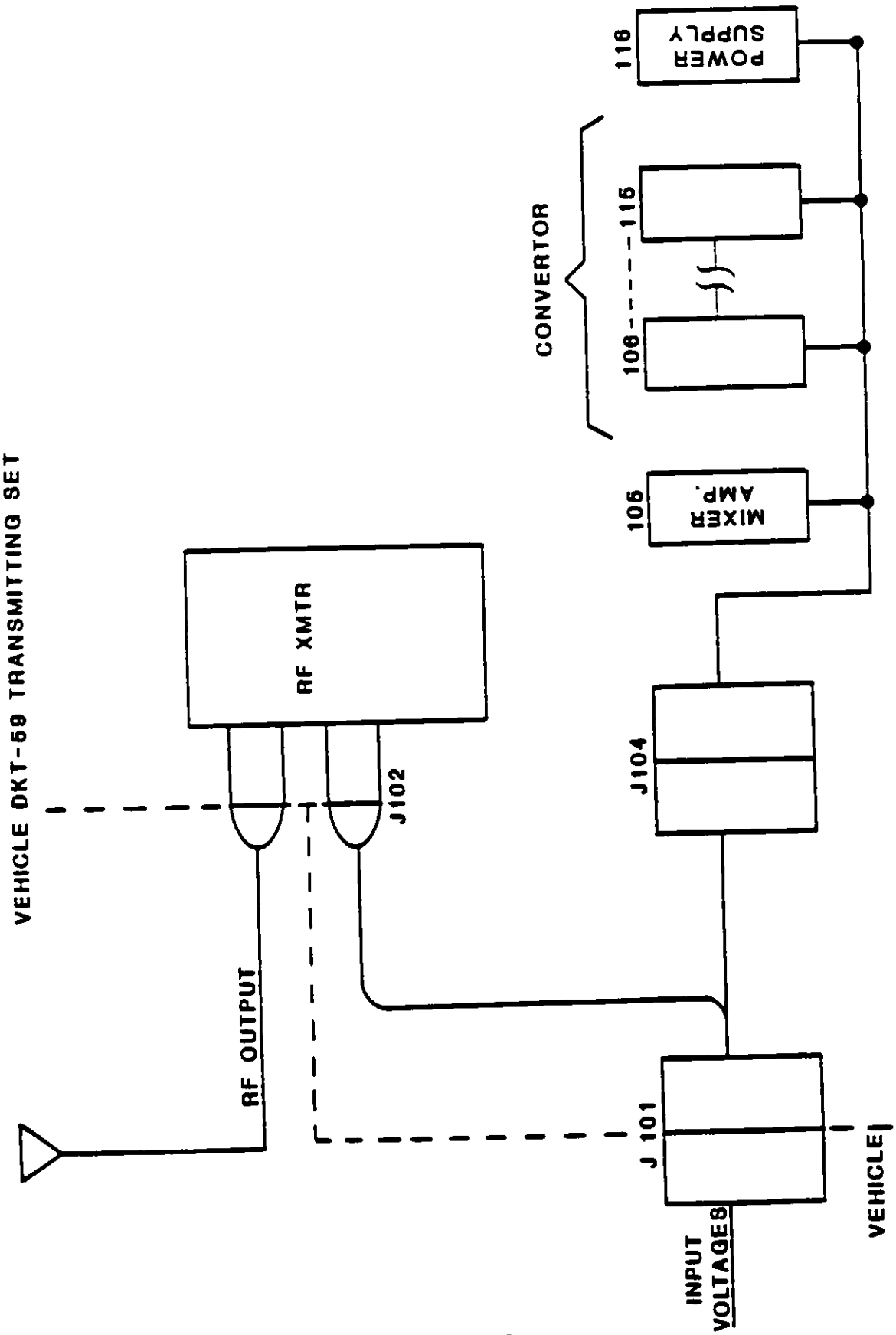


FIGURE 1. Interface definition diagram, telemetric data transmitting set

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b. Transient surge from +40 to -40 Vdc

c. Maximum current shall be not greater than 2.15 amperes

3.2.1.1.1 Under-voltage protection. The transmitting set shall not be damaged by primary power voltages less than 20.0 Vdc, although specified performance is not required during low-voltage periods. The transmitting set shall automatically resume specified operation when the voltage returns within the specified limits (see 4.6.2.3).

3.2.1.2. Input characteristics. The transmitting set shall have the following signal input characteristics:

a. Input impedance: Shall be not less than 500 Kilohms (K ohms) (see 4.6.2.4).

b. Input signal range: 0.0 to 5.0 Vdc (with mode adjust input grounded) or -2.5 to +2.5 Vdc (with mode adjust input ungrounded) (see 4.6.2.4).

3.2.2 Modulation characteristics.

3.2.2.1 Modulation compatibility. The transmitting set shall be capable of operating within the requirements of this specification and IRIG Document 106-80 (see 4.6.2.5).

3.2.3 Carrier deviation.

3.2.3.1 Modulation sensitivity. The modulation sensitivity shall produce a 100 KHz deviation ±10 percent with 2.5 Vdc input applied to each channel (see 4.6.2.6).

3.2.3.4 Spurious amplitude modulation. The peak-to-peak amplitude modulation of the carrier output signal shall be not more than 5 percent of the carrier root mean square (rms) amplitude value with any input signals as specified in 3.2.1.2b.

3.2.3.5 Spurious outputs. All spurious and harmonic outputs measured in the transmission line between the transmitting output and the load shall be greater than 65 dB down in power from the carrier level to within 500 KHz of the carrier, and shall be greater than 40 dB below the carrier level from 500 KHz to 320 KHz of the carrier. This requirement shall apply to spurious components falling within the modulation bandwidth of the transmitting set, as well as those harmonic and spurious components outside the modulation bandwidth of the transmitting set from 0.150 MHz to 10,000 MHz. Signals

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conducted by power leads or radiated directly from units and cables shall be within the limits specified in MIL-STD-461, Test Requirements CE01, CE03, CE07, CS01, CS02, CS06, RE02, RS02 and RS03 (see 3.4.3 and 4.6.2.8).

3.2.3.6 Temperature stability. The transmitting set shall be capable of operating from -40° degrees Centigrade (°C) to +71°C with no major or permanent degradation in excess of the limits specified herein (see 4.6.2.9).

3.2.3.7 Interchangeability. All assembly and subassembly level items of hardware that are subject to replacement and have the same part number shall be electrically, physically, and functionally interchangeable. Electrical component parts shall conform to the requirements specified in MIL-STD-965. The transmitting set shall meet the interchangeability requirements of MIL-STD-454, Requirement 7, and the electrical and mechanical interface requirements specified herein (see 4.6.1.6 and 4.6.2.10).

3.2.4 Reliability. The reliability of the transmitting set shall be not less than 200 hours mean-time-between-failures (MTBF). The minimum acceptable MTBF shall be not less than 100 hours when tested and accepted under the conditions of 4.4.1 (see 4.6.4).

3.2.5 Maintainability. The maintainability program, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in MIL-STD-2078 (see 4.6.1.10).

3.2.6 Environmental conditions.

3.2.6.1 Operational. The transmitting set shall meet all requirements of 3.2.1 and 3.2.4 prior to, during, and after exposure to all the following environmental conditions, separately or any combination thereof. The transmitting set shall not exhibit any temporary or permanent degradation in excess of the limits specified herein as the result of exposure to any of the environments in a. through f. of Table I (see 4.6.3.1).

3.2.6.2 Non-operational. The transmitting set shall meet all requirements of 3.2.4 and 3.2.1 after exposure to the environmental conditions c, h, and g of Table I, separately or any combination thereof. The transmitting set shall not exhibit any temporary or permanent degradation in excess of limits specified herein as the result of exposure to any of the environments specified herein (see 4.6.3.2).

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TABLE I. Environmental conditions.

PARAMETER	OPERATIONAL	NON-OPERATIONAL
a. Pressure	70,000 ft (equivalent pressure)	-----
b. High Temperature	+71°C	-----
c. Low Temperature	-40°C	-40°C
d. Acceleration	40g, 3 axes	-----
e. Random Vibration	20-2000 Hz, $.2g^2/Hz$	-----
f. Shock	65g, 5ms, 1/2 sine	-----
g. Thermal Shock	-----	-55°C to +85°C
h. Moisture Resistance	-----	95% relative humidity

3.2.7 Storage life. The storage life of the transmitting set shall be such that it can be stored for three years in the shipping or storage container and then meet the requirements specified herein.

3.3 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3.1 Visual examinations. Test samples shall be subjected to visual examination to verify conformance to the physical and interchangeability requirements of this specification and the contract (see 4.6.1).

3.4 Design and construction. Design and construction shall be in accordance with Drawing 1356AS309 and requirements specified herein (see 3.4.1 and 4.6.1.9).

3.4.1 Materials, processes, and parts. Fulfillment of major design objectives for the intended application specified herein shall be the prime consideration in the selection of materials, processes, and parts. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein (see 4.6.1.2).

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b. Other materials, processes, and parts requirements shall conform to MIL-E-5400 (see 4.6.1.2).

c. Parts shall be selected using the following order of preference (see 4.6.1.2).

(1) Parts listed on a Qualified Parts List (QPL) with two or more sources.

(2) Parts listed on a QPL with only one source.

(3) Commercial parts with two or more sources.

(4) Commercial parts with only one source.

3.4.1.1 Standard parts. Parts specified in 3.4.1c (1) and (2) are standard parts and do not require approval from the procuring activity. Parts selected for use from 3.4.1c (3) and (4) are considered nonstandard and require approval. Nonstandard parts approval requests shall be submitted on DD Form 2052 in accordance with the contract (see 6.2.2).

3.4.1.2 Commercial parts. Commercial parts specified in 3.4.1c (3) and (4) shall be readily available, nationally catalogued, off-the-shelf items.

3.4.1.3 Nonstandard parts. Use of nonstandard parts in first article or production units prior to Government approval is at the sole risk of the contractor and subject to replacement if the Government does not approve the proposed nonstandard part.

3.4.2 Electromagnetic interference. The transmitting set shall meet the electromagnetic interference (EMI) requirements of MIL-STD-461, tests CE01, CE03, CE07, CS01, CS02, CS06, RE02, RS02, and RS03.

3.4.3 Nameplate and marking. The nameplate shall be as specified in Drawing 1356AS309. Interchangeable parts and assemblies shall be legibly marked. All markings shall be as specified in MIL-STD-130 (see 4.6.1.3).

3.4.4 Serial numbers. Serial number assignments shall be as specified in MIL-N-18307.

3.4.5 Safety. The transmitting set shall conform to the safety requirements of MIL-STD-454, Requirement 1 (see 4.6.1.7).

3.4.6 Human engineering. Unless otherwise specified herein, all controls and adjustments shall be set during production and initial testing, and shall be inaccessible in the field (see 4.6.1.8)

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3.5 Soldering. Soldering practices shall be in accordance with MIL-STD-454, Requirement 5 (see 4.6.1.4).

3.6 Workmanship. Fabrication and workmanshp practices shall be in accordance with MIL-STD-454, Requirement 9 (see 4.6.1.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to monitor any of the inspections or to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall assure product conformance to the requirements, inspections and tests specified herein. The contractor's Quality Assurance Program shall be planned and conducted in a manner to achieve levels of quality and reliability commensurate with the intended application specified herein.

4.1.1.1 Inspection. The contractor's inspection system shall conform to MIL-STD-883 for both the commercial off-the-shelf microelectronics and new contractor-designed (peculiar) microelectronics.

4.1.1.2 Test equipment and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.1.2 Test plans and procedures. Detailed test plans and procedures in accordance with MIL-T-18303 shall be prepared by the contractor (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.1.3 Test data. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare test data showing quantita-

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tive results for all first article and quality conformance inspections (see 6.2.2) for each transmitting set offered for acceptance. This report shall also identify all rejections from tests.

4.1.3.1 Electromagnetic interference data. The contractor shall prepare a electromagnetic interference test report in accordance with MIL-STD-831 (see 6.2.2).

4.1.4 Failure reports. All test failures shall be documented in a failure report, identifying the cause of failure and the proposed corrective action for each failure (see 6.2.2).

4.1.5 Reliability Reports. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare reliability test data, including failure reports, in accordance with MIL-STD-781 (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

(1) Visual examination (see 4.6.1)

(2) Performance test (see 4.6.2)

(3) Environmental test (see 4.6.3)

(4) Reliability test (see 4.6.4)

b. Quality conformance inspection (see 4.5)

(1) Individual test (see 4.5.2)

(2) Periodic production test (see 4.5.3)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.3.3.

4.3.1 Maintenance and calibration. All test equipment shall be calibrated and maintained in accordance with MIL-STD-45662. Certification to this effect shall accompany all reports of first article and quality conformance tests (see 6.2.2).

4.3.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy within 20 percent of the tolerance

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for the variable to be measured. In those cases where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within five percent of that requirement.

4.3.3 Test environment. Unless otherwise specified in the applicable test procedure, all tests and inspections shall be conducted under the following environmental conditions:

a. Temperature (ambient): +23 ±10°C

b. Altitude: elevation of test facility

c. Relative humidity: 50 ±30 percent

4.3.4 Tolerances on environmental test conditions. The allowable tolerances for environmental test conditions shall be in accordance with MIL-T-5422.

4.4 First article inspections. When specified in the contract (see 6.2.1), first article inspections shall be conducted by the contractor on a transmitting set representative of the production transmitting sets to be supplied under the contract. The first article sample shall be subjected to the examinations and tests of Table II and shall be performed in the sequence specified therein.

4.4.1 Acceptance criteria. Failure of any transmitting set to pass any first article test shall result in discontinuance of testing and rejection of the first article sample. The procuring activity reserves the right to require an additional sample and test if any transmitting set fails to meet any requirement of this specification.

4.4.1.1 Post-tests. A transmitting set which has previously failed first article inspection shall be retested in accordance with 4.6.2 prior to their resubmission for first article acceptance.

4.4.2 First article failure analysis and corrective action. The procuring activity shall be notified of any first aticle test failure and its probable cause (see 4.1.4 and 6.2.2).

4.5 Quality conformance inspections. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspections. All inspection and testing shall be subject to monitoring by the Government inspector. Acceptance or approval of material during the

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TABLE II. Individual tests.

Examination or Test*	Requirement Paragraph	Method Paragraph		
		First Article	Quality Conformance	
			Individual	Periodic Production
*Visual examination	3.3.1	4.6.1	4.6.1	4.6.1
Performance tests				
External power	3.2.1.1	4.6.2.2	4.6.2.2	4.6.2.2
Under-voltage protection	3.2.1.1.1	4.6.2.3	4.6.2.3	4.6.2.3
Input characteristics	3.2.1.2	4.6.2.4	4.6.2.4	4.6.2.4
Modulation compatibility	3.2.2.1	4.6.2.10		4.6.2.10
Modulation sensitivity	3.2.3.1	4.6.2.12		4.6.2.12
Spurious amplitude modulation	3.2.3.4	4.6.2.15		4.6.2.15
Spurious outputs	3.2.3.5	4.6.2.16		4.6.2.16
Temperature stability	3.2.3.6	4.6.2.17	4.6.2.17	4.6.2.17
Interchangeability	3.2.3.7	4.6.2.18	4.6.2.18	4.6.2.18
Electromagnetic interference	3.4.2	4.6.2.1	4.6.2.1	4.6.2.1
Environmental tests	3.2.6	4.6.3	4.6.3	4.6.3
Reliability tests	3.2.4	4.6.4	4.6.4	

*All examinations and tests shall be performed in top to bottom sequence.

course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspections shall consist of the following:

a. Individual tests (see 4.5.2)

b. Periodic production tests (see 4.5.3)

4.5.1 Quality conformance acceptance criteria. Any transmitting set that fails to pass any individual test shall be rejected.

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4.5.2 Individual tests. The individual tests shall be conducted on each transmitting set offered for acceptance and shall consist of the tests in Table II.

4.5.3 Periodic production tests. After the first 50 transmitter sets are produced, a periodic sample consisting of one unit shall be selected at random from the first 10 transmitting sets of every 50 produced thereafter. This sample shall be selected from transmitting sets which have previously passed the individual tests of 4.5.2. The sample shall be subjected to all the examinations and tests specified in Table II.

4.5.3.1 Periodic production sample rejection. If the periodic production sample fails to meet the requirement of any examination or test specified herein, the sample shall be rejected. Before any further production lots or transmitting sets may be submitted for acceptance, a detailed report shall be prepared (see 6.2.2) describing the rejection and action taken to prevent recurrence of the defect which caused the failure. Further production lots will not be considered for acceptance until the periodic production sample has been approved.

4.6 Examination and test methods. Details of methods and procedures shall be in accordance with the approved test plan and procedures (see 4.1.2).

4.6.1 Visual examination.

4.6.1.1 Physical characteristics. The transmitting set shall be inspected visually for conformance to 3.1.2

4.6.1.2 Materials, processes, and parts. The transmitting set shall be examined visually for conformance to 3.4.1.

4.6.1.3 Nameplates and product marking. The transmitting set shall be inspected for conformance to 3.4.3.

4.6.1.4 Soldering. The transmitting set shall be examined for conformance to 3.5.

4.6.1.5 Workmanship. The transmitting set shall be examined for conformance to 3.6.

4.6.1.6 Interchangeability. The transmitting set shall be examined to verify physical interchangeability conformance to 3.2.3.7.

4.6.1.7 Safety. The transmitting set shall be examined to verify conformance to the safety requirements of 3.4.5.

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4.6.1.8 Human engineering. The transmitting set shall be examined to verify conformance to the human engineering requirements of 3.4.6.

4.6.1.9 Design and construction. The transmitting set shall be examined to verify conformance to the requirements of 3.4.

4.6.1.10 Maintainability. The transmitting set maintainability program, construction, packaging and access to test points shall be examined to verify conformance to the requirements of 3.2.5.

4.6.2 Performance tests. Performance tests shall be performed as specified herein.

4.6.2.1 Electromagnetic interference. The transmitting set sections shall be tested for conformance to 3.4.2 and in accordance with MIL-STD-462.

4.6.2.2 External power. External power shall be measured to verify conformance to the requirements of 3.2.1.1 prior to conducting the tests specified in 4.6.2.3 through 4.6.2.8 to ensure the accuracy and validity of those tests.

4.6.2.3 Under-voltage protection. The transmitting set shall be tested for conformance to 3.2.1.1.1 to verify that it will not be damaged by primary power voltages less than 20.0 Vdc, and that the transmitting set will automatically resume specified operation when the voltage returns within the specified limits.

4.6.2.4 Input characteristics. The input signal characteristics to the transmitting set shall be measured to verify conformance to the requirements of 3.2.1.2 prior to conducting the tests specified in 4.6.2.4 through 4.6.2.10 to ensure the accuracy and validity of those tests.

4.6.2.5 Modulation compatibility. The transmitting set modulation compatibility shall be tested for conformance to the requirements of 3.2.2.1.

4.6.2.6 Modulation sensitivity. The transmitting set modulation sensitivity shall be tested for conformance to the requirements of 3.2.3.1.

4.6.2.7 Spurious amplitude modulation. The transmitting set shall be tested to verify conformance to the requirements of 3.2.3.4.

4.6.2.8 Spurious outputs. The transmitting set shall be tested to verify conformance to the requirements of 3.2.3.5.

4.6.2.9 Temperature stability. The transmitting set temperature stability shall be measured at -40°C, +25°C, and +71°C and shall conform to the requirements of 3.2.3.6.

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4.6.2.10 Interchangeability. The transmitting set shall be tested to verify electrical and operational interchangeability conformance to 3.2.3.7.

4.6.3 Environmental tests. Environmental tests shall be performed as specified herein. When tests are conducted at high or low temperatures, test transmitting sets shall be stabilized at the specified temperature for not less than two hours prior to testing. If the transmitting sets are removed from the chamber for any portion of the tests, such tests shall be completed within 10 minutes of removal from the chamber. If the tests are not completed within the 10 minute period, the transmitting sets shall be returned to the temperature chamber for not less than two hours before continuing the tests. The test transmitting sets shall stabilize at room ambient conditions prior to performance of post tests and measurements.

4.6.3.1 Operational environmental tests. The transmitting set performance shall be tested as specified in 4.6.3.1.1 through 4.6.3.1.6 and conform to the requirements of 3.2.1.1 through 3.2.3.7 during and after exposure to the environmental conditions specified in Table I, a through f.

4.6.3.1.1 Reduced pressure. The transmitting set shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.2 High temperature. The transmitting set shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.3 Low temperature. The transmitting set shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.4 Acceleration. The transmitting sets shall be tested in accordance with 3.2.6.1 and Requirement d. of Table I.

4.6.3.1.5 Random vibration. The transmitting set shall be tested with a spectral power density of 0.2 g^2/Hz from 20 to 2000 Hz for 2 hours in each of three mutually perpendicular axes.

4.6.3.1.6 Shock. The transmitting sets shall be tested in accordance with MIL-T-5422, Shock Test.

4.6.3.2 Nonoperational environmental tests. The transmitting set's performance shall be tested as specified in 4.6.3.2.1 through 4.6.3.2.3 and conform to the requirements of 3.2 after exposure to the environmental conditions specified in Table I, c, g, and h.

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4.6.3.2.1 Low temperature. The transmitting set shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 equipment Class 2 at -40°C for 48 hours.

4.6.3.2.2 Thermal shock. The transmitting set shall be tested in accordance with MIL-T-5422, Temperature Shock Test.

4.6.3.2.3 Moisture resistance. The transmitting set shall be tested in accordance with MIL-T-5422, Humidity Test.

4.6.4 Reliability test. The transmitting set shall be tested as specified in the Probability Ratio Sequential Test (PRST) Plan VIIIC, Figure C-8, Appendix C of MIL-STD-781 to demonstrate compliance with 3.2.4.

4.6.4.1 Reliability acceptance criteria. The Government inspector shall be notified any time the total test hours and test failures plotted in accordance with the PRST Acceptance and Reject criteria for Test Plan VIIIC of MIL-STD-781 reflect a "reject" situation during the conduct of the contractor's approved production test procedures. The procuring activity reserves the right to stop the acceptance of transmitting sets at any time a reject situation exists pending a review of the contractor's efforts to improve the transmitting sets, their components, parts and the workmanship.

4.7 Production transmitting sets. Transmitting sets supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article. Each transmitting set shall be capable of successfully passing the same tests as imposed on the first article. Evidence of noncompliance with the above shall constitute cause for rejection, and for transmitting sets already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections.

4.8 Packaging inspection. The preservation, packaging, packing and marking of the transmitting set shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the level of preservation, packaging, and packing shall be as specified in MIL-E-17555 and MIL-STD-794, Level C. The method of preservation for Level C shall be determined in accordance with the selection chart in Appendix D specified in MIL-STD-794 and the method of packaging shall be as specified in MIL-P-116.

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5.2 Marking

5.2.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.2.2 Normal marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3 Spare parts. When specified (see 6.2.1), spare parts shall be preserved and packaged to the same specifications as a complete transmitting set except that weight of spare parts will determine selection of style of the box or container.

6. NOTES

6.1 Intended use. The AN/DKT-59 Telemetric Data Transmitting Set is intended for use in the BQM-74C, BQM-34A, BQM-34E, and other aircraft targets to provide pre-flight and mission flight data for technical test and evaluation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and date of this specification.

b. Number of transmitting sets to be procured.

c. First article inspection if different from 4.4.

d. Test plans and procedures if different from 4.1.2.

e. Facility assigned to perform inspections (see 4.1).

f. Test data and reports if different from 4.1.3.

g. Acceptance criteria if different from 4.4.1 and 4.5.1.

h. Government loaned property if different from 3.1.5.

i. Inspection conditions if different from 4.3.

j. Test equipment accuracy if different from 4.3.2.

k. Level of packing and packaging if different from 5.1.

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l. Special marking (see 5.2.1).

m. Spare parts preservation and packaging (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specificaiton are cited in the following paragraphs.

Paragraph Number	Data requirements	Applicable DID Number	Options
4.1.2	Procedures, Test	UDI-T-21347	----
4.1.3	Reports, Test	DI-T-2072	----
4.1.3.1	Electromagnetic Interference Test Report	DI-R-7062	----
4.1.4	Report, Failed Item Analysis	DI-R-7039	----
4.1.5	Reports, Reliability Test and Demonstration	DI-R-7034	----
4.3.1	Equipment Calibration Procedures	DI-R-7065	----
4.4.2	Failure Analysis and Corrective Action Report (First Article)	DI-R-5299	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the first article sample.

Preparing activity:
Navy - AS
(Project No. 5815-N115)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85650(AS)	Transmitting Set, Telemetric Data AN/DKT-59

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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